Student Engineer Design Portfolio

Due: as per exams **Weight:** 15% **Submitter:** Individual

Overview

A design portfolio is a document that showcases you as a designer and the products of your design activities for an external audience, often used for job applications in design or to solicit clients. It is a curated collection of your work assembled in order to demonstrate your skills and experiences, and sometimes also to provide evidence that you practice according to your position in context. A design portfolio often includes summarized descriptions of the products of your design work, like the one-pager you will create for your Praxis II Showcase. The design portfolio is a document that you can build upon into your future career as you encounter more design experiences. In Praxis II, your design portfolio will provide evidence that you meet all of the high-level learning outcomes associated with this course, which are that you:

1. Engage in quality, credible engineering design;
2. Express engineering designs and arguments using modes appropriate to engineering;
3. Work effectively as a member of a team.

In Praxis II, you will create a design portfolio that is suited to your current position as a student engineer. This will be a hybrid of a typical professional design portfolio, which showcases the products of your design work, and an engineering handbook. A handbook is a reference intended for individuals who are already familiar with the topic area and, in the case of a personal handbook, want to keep a record of key parts of their process for future reference. For example, you can use your handbook:

- as a checklist to ensure you have rigorously practiced engineering design,
- to remind yourself of which tools you use, and why/how you have used them in the past, including cases where you should avoid using a tool,
- to remind yourself how to perform a specific task or select among different tools or models, and/or
- to browse when you're exploring possible approaches to addressing a design opportunity or a given phase of the design process.

We are asking you to use your design work—whether notes, prototypes, final concepts, or process components—to create a curated record of concepts, tools, models, and frameworks (CTMFs) used in developing your design practice and of the value of these tools to your design work and team function. As with any good handbook, you will need to explain each CTMF thoroughly enough that you can use it later (perhaps after you've forgotten most of first year), know when and why to use it or not use it, and what you should expect to gain from it. You will select your best evidence to demonstrate your effective use (or not) of the tools or models, and you will reflect upon your work in a way that provides an honest appraisal of your accomplishments.

Stakeholders

Your portfolio has the following main stakeholders:

You: The goal of this assignment is for you to create something that is a professional representation of your design work, and will be useful to you as a personalized reference about your actual process in your future engineering design activities. These activities may be in the classroom (e.g., ESC204 and your capstone design course), as part of a student club (e.g. UTAT, BlueSky, etc.), or on the job (e.g. summer employment, your professional experience year, or after you graduate).

Praxis II teaching team: The teaching team's primary interest is to assess your articulation of your (developing) engineering design practice, your understanding of selected design CTMFs you can and have used in your design practice, and your understanding of your overall process and use of CTMFs. They would also like to be able to complete this assessment in a reasonable amount of time.

Other design courses (e.g. ESC204, capstone design, etc.): The teaching teams in other design courses would like to focus their teaching on new content, not on reminding you about materials that you have already been taught. They also expect you to be able to manage your own design activities in a rigorous and comprehensive way.

Other engineering designers: Depending on your format, you could end up deciding to share your position on engineering design and associated methods with other engineering designers. This would be especially true if you were trying to collectively determine a way to collaborate on a design project in a future team.

Portfolio components

This assessment is built on understanding yourself as engineering designer. It invites you to revisit your position statement to consider not only how it may have changed over the few months, but also how the tools and evidence that you present will help you to strengthen your strengths and develop in areas of growth. We encourage you to use your position as a way to frame your portfolio: it may be helpful in thinking about how to structure your portfolio overall, or may influence the types of descriptions you make about the tools you use(d). Your portfolio must do three main tasks, which correspond to three main components, as shown in Figure 1. These three tasks are as follows:

1. **Establish your position in context**. This should iterate on your original position statement; your position will likely have evolved. Introduce yourself and your portfolio in terms of your position as an engineering designer and how you approach engineering design. In doing so, you should be establishing the basis for your portfolio's organization, the set of CTMFs you discuss, and for your assessments of those materials. Your original position statement may or may not fit easily into the portfolio. You may choose to revise your position statement to tailor it to this portfolio, or you may simply add an annotation[1] that integrates it into your portfolio.

[1] This is primarily intended to avoid an unreasonable amount of work in revising a position statement that is represented in a format that is not easy to edit. You are welcome to talk to the teaching team about this.

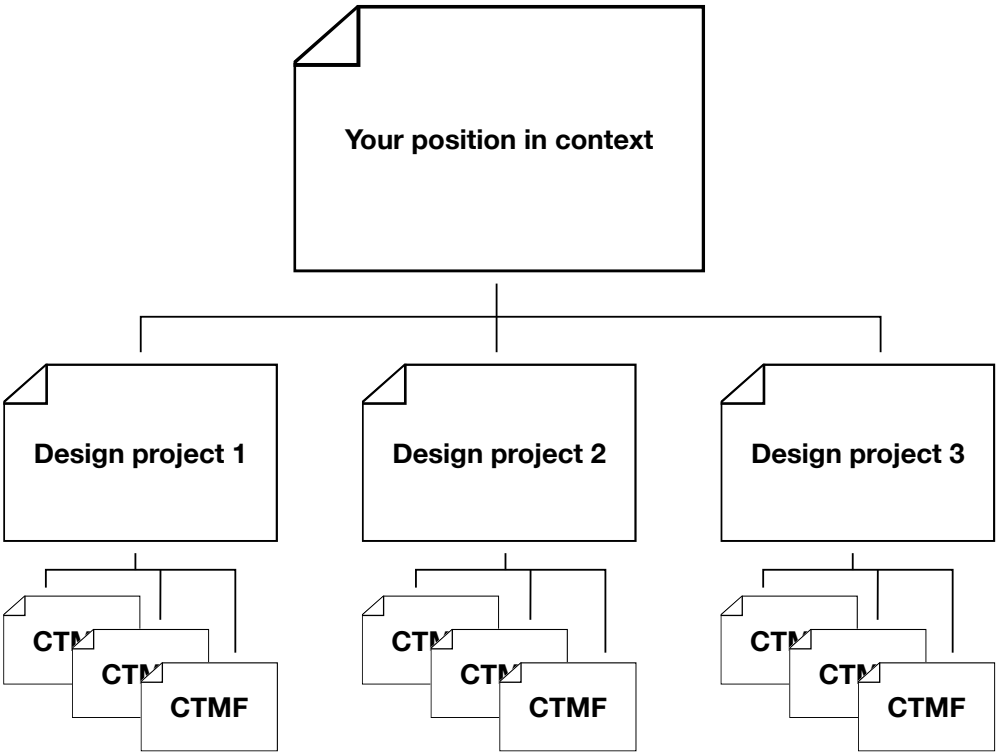

Figure 1. A graphical representation of the components of your design portfolio and their relationship to one another. This representation shows the minimum components required. The relative size of the components is a layout choice and is not intended to suggest relative amounts of content for different components.

2. **Present the products of your design experiences and explain and interpret relevant aspects of the process**. You should summarize your design experiences in a creative and informative manner. The summary should give enough information that someone who is not familiar with the project or context can understand what was done and why. It should emphasize how your position was enacted in your designs and should balance documentation of the process and its resultant product.

3. **Select and assess design concepts, tools, models, and frameworks** (CTMFs[2]) that you have used in your process and want to capture for future use (or ensure you do not get trapped using them again). Your portfolio should:

   - Provide evidence for the use of design CTMFs that span at the entire Frame • Diverge • Converge • Represent process, and are organized in a manner that is consistent with your position and practice.

   - Evaluate the effectiveness of the selected CTMFs in your design practice. An assessment could critique the CTMF, or consider when, where, and why you should use it, or not, and what can be gained by using it or avoiding it.

[2] This assignment uses these related terms to avoid anchoring you to the commonly understood engineering design tools (e.g. measurement matrix; Pugh chart; pairwise comparison; etc.) The additional terms are intended to suggest that other design, communication, and teamwork concepts can also be included in your portfolio. **Precision in distinguishing between a concept, tool, model, and framework is not necessary.**

Assignment requirements

Requirements language in this and following sections are to be interpreted as described in RFC 2119.

Objectives

The objectives of the Design Portfolio assignment are as follows:

1. Represent a personal position on engineering design that enables you to effectively manage your engineering design activities and teams.
2. Document your engineering design work to date as a student engineer.
3. Document preferred engineering design CTMFs[3] (from the design literature, from lectures in Praxis I and II, from practice, etc.) that are relevant and applicable to your position on engineering design.
4. Demonstrate having applied or having attempted to apply such CTMFs in your design activities.
5. Reflect on your application of the CTMFs and document your assessment of their usefulness and value in the context of your design activities, skills, teamwork, and values.
6. Demonstrate your engineering communication skills. Clearly communicate your content to audiences that include the teaching team, teammates, and your future self. In addition to the sections denoted here, such communication should include, at minimum,
    a. an introduction that will prepare a reader (which could include you 8 months later) to understand the purpose and navigate the document effectively;
    b. a logical organization and formatting that a viewer can follow easily, with flexible navigation suitable to the content structure;
    c. concise writing, such that each word or sentence is adding meaning and value;
    d. consistent and well-chosen visual elements to illustrate concepts or design work;
    e. appropriate use of conventions such as table of contents, headings, paragraphs, bullet lists, and captions for visual elements.

[3]These CTMFs should enable you to practice engineering across an entire engineering design project from opportunity finding to verification and validation of a recommended design.

Requirements

Your Student Engineer Design Portfolio:

1. **Shall** be submitted to Quercus as a single PDF document. The use of internal PDF links is encouraged but is not required for PDF submissions. You may also choose to build your portfolio as a website. If you choose to build a website, you should submit a PDF that contains a link to the website and provides screenshots of the main pages, taken before the submission deadline.[4]
2. **Shall** include a reference list at the end of the PDF document in the Praxis "embedded extract" form using a standard format (e.g. APA, IEEE). If you build a website, you should include references and an appropriate reference list on your website, and only include your source extracts in the PDF you submit to Quercus.
3. **Shall** include a statement of your position as an engineering designer (note: this should be an iteration on the position statement you created in January, and should reflect any updates to your understanding and approach since then).
4. **Shall** include one-page summaries[5] of your primary design activities completed since September 2025 (Praxis I project, CIV102 Bridge Project, Praxis II project). May include one-page summaries of any other relevant design activities since September 2025.
5. For each design project, **shall** include an annotation that explains and interprets the design project (including the process) in context.
6. **Shall** only include design projects done since September 2025. May include evidence from prior design activities as part of your position in context.[6]
7. For each of the design projects in your portfolio, **shall** include at least three (≥3) distinct CTMFs that were used in or are relevant for that design project, for a total of at least nine (≥9) distinct CTMFs.
   a. You may connect any CTMF to more than one design project, if there is reason to do so, but you must have at least three (≥3) distinct CTMFs for each project.
8. **Shall** include CTMFs that cover all strands of Frame • Diverge • Converge • Represent
9. **Should** include only CTMFs for which you can provide evidence of your use in an engineering project context, or that you can credibly connect to an engineering project by retroactively analyzing it.[7]
10. **Shall** give credit for design work performed by teammates (as per the University of Toronto Code of Academic Behaviour and the Professional Engineers Ontario Code of Ethics).

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[4] You are responsible for ensuring that a website submission is accessible and stable during the grading period. If there is an issue with the website when your assessor wants to grade it, they may use your PDF version instead.

[5] To be consistent with the Praxis II "one-pager" you will create for Showcase, these summaries may actually contain two pages' worth of content, since the Showcase one-pager is limited to one **double-sided** US-letter-sized page.

[6] If you want to use something from before September 2024, it would be a good idea to check with the teaching team.

[7] This may be the case for your CIV102 bridge, if your team did not actively apply Praxis thinking during that design process. You should be able to make connections between your bridge design process and relevant Praxis CTMFs.

Evaluation criteria

Please refer the assignments associated independent assessment tool for the metrics associated with these characteristics of evaluation. For all characteristics, the criteria "more", "higher", or "greater" are preferred.

1. The clarity of your position in context and its effectiveness in representing your approach to engineering design.
2. The validity of your position as demonstrated through evidence from and explanation/interpretation of your projects.
3. The appropriateness of the selection of evidence from design projects to reflect how you practice engineering design according to your position.
4. The utility and clarity of your explanations and annotations in providing context for and interpreting your design projects, and in linking your evidence to your position and design products.
5. The appropriateness of the selection of CTMFs based on actual experience in using them and/or their relevance to your design work, and based on a legitimate ability to assess them from having used them.
6. The utility and clarity of your explanations and annotations in providing context for and evaluating your use of CTMFs in your design process.
7. The ease of navigation through the portfolio, including the ease with which a reader can find supporting evidence for the claims in your document.
8. The clarity, correctness, and conciseness of written prose.
9. The value, quality, and integration of visual elements in your portfolio.[8]

[8] Since some of these images may be taken from whiteboard, virtual whiteboard, etc. the quality of the image itself is less important than the quality of the content represented. However, if the image is difficult to read, you should make use of annotations, captions, and any other relevant means to facilitate the viewers' understanding. In some cases redrawing the contents of the visual is the best approach.